

Priyanka Verma

PH.D. CANDIDATE, UNIVERSITY OF TORONTO

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As an interdisciplinary researcher, I design, deploy, and evaluate data tools and algorithmic systems, with a particularly focus on financial practices and technologies. Experienced building and deploying end-to-end technical systems across government, non-profit, and tech sectors, often in cross-disciplinary collaborations.

EDUCATION

Ph.D. Candidate, Information Technology

2022 - 2026

UNIVERSITY OF TORONTO, CANADA

- Technical AI Safety Fellow at University of Toronto- Engaging in the weekly reading group for AI Alignment track at Trajectory Labs (Spring 26)
- Lecturer for a senior undergraduate UX methods course: designed new modules, assignments about AI
- Spearheaded a mentorship program for 85+ doctoral applicants from underrepresented groups
- Presented research at conferences (CHI, NeurIPS), industry (EQ Bank), non-profits (CCRI, Co-social)

M.Sc. (Hons.) Economics and B.E. (Hons.) Computer Science

2016- 2021

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE (BITS), PILANI, INDIA

EXPERIENCE

Software Developer, Oracle

July 2021- Aug 2022

BANGALORE, INDIA

- Worked on the organization-wide cloud logging and monitoring agent Key contributions:
 - contributed to [open-source logging output plugin](#) to support Debian OS
 - enabled ingestion, aggregation of custom logs by developing 18+ parser plugins
 - reduced the security risk score from 9.2 to 6.4 of the agent by tuning systemd configurations

Research & Software Development, CGNet Swara

Feb 2021- June 2021

DR. BILL THIES, MICROSOFT RESEARCH INDIA

[\[Thesis\]](#) [\[Demo\]](#)

- *CGNet Swara provides a voice-based citizen journalism platform to rural communities in Central India*
- Implemented an **interactive WhatsApp chatbot** for audio-, video-, image-based journalism posts
- Developed customised features to allow users to request content by using contact card and location features
- Successful adoption of chatbot with 500+ reports from low-literate users within 9 weeks; continued usage

Research Fellow, Indiana University Bloomington, USA

Aug 2020- Jan 2021

DR. SAMEER PATIL, UNIVERSITY OF UTAH

[\[Thesis\]](#)

- 1 of 10 recipients of Global Talent Attraction Program scholarship by Indiana University Bloomington
- Designed and analysed a survey for 200+ participants to study privacy aspects of smartphone notifications
- Conducted **mixed-method** analysis (non-parametric statistical tests, thematic analysis, visualizations)
- Recommended design recommendations for smartphone UIs to prevent privacy concerns

R&D Intern, BISAG-N, Govt. of Gujarat

May 2018 - July 2018

SENIOR SCIENTIST MANOJ PANDYA

[\[Repository\]](#)

- *BISAG-N is a scientific society under Govt. of India focusing on e-Governance, Geo-informatics projects*
- Implemented a smart app (Elrefaei et al. 2015) to automate electricity meter reading and bill generation
- Extracted meter readings from videos in real-time using matching and feature extraction algorithms
- Received **Innovation Challenge Award** from Dept. of Science and Technology, Govt. of Gujarat

PROJECTS

Fairness in Credit Scoring Models

2024

DR. SHION GUHA, UNIVERSITY OF TORONTO

[\[Documentation\]](#)

- Evaluated credit scoring models for fairness and disparities quantitatively and qualitatively
- Trained ML model on German-Credit data and computed statistical discrimination metrics

Bayesian Inference Analysis of Credit Card Default Payment

2023

DR. MONICA ALEXANDER, STATISTICAL SCIENCES, UNIVERSITY OF TORONTO

[\[Documentation\]](#)

- Applied Bayesian statistics frameworks in R to investigate real customers' credit card defaulting behavior
- Implemented two hierarchical logistic regression models, computational and posterior predictive checks

Analysing Impact of COVID-19 on US Child Welfare

2022

DR. ASHTON ANDERSON, COMPUTER SCIENCE, UNIVERSITY OF TORONTO

[\[Documentation\]](#)

- Analysed over 12,566 casenotes from a US Child Welfare (CW) Agency to investigate impact of COVID-19
- Conducted topic modelling, sentiment analysis, and open coding on CW Agency casenotes | [Poster](#)

Multivariate Time Series Analysis and Applications

2019

PROF NAVNEET GOYAL, ADVANCED DATA ANALYTICS LAB- BITS PILANI

[\[Documentation\]](#)

- Investigated time series similarity measures, clustering and classification methods and Granger causality
- Analysed earthquakes time series data; Achieved 87.76% classification accuracy

Image Caption Generation

2019

PROF NAVNEET GOYAL, ADVANCED DATA ANALYTICS LAB- BITS PILANI

[\[Repository\]](#)

- Used 'merge-model' architecture by Marc Tanti, et al. (2017) in **Keras** for :
 - feature extraction from images using VGG model
 - textual descriptions generation using **word embeddings** & **LSTM** neural network
 - predictions using decoder model over the entire output vocabulary for next word
- Achieved BLEU score of .528 for Flickr-8k dataset

SELECT PUBLICATIONS

Full list: [Google Scholar](#)

- **Priyanka Verma** and Balagopal Unnikrishnan. When AI Democratizes Exploitation: LLM-Assisted Strategic Manipulation of Fair Division Algorithms. arXiv:2511.14722. **NeurIPS** 2025 workshop on Algorithmic Collective Action
- **Priyanka Verma**, Christoph Becker, and Samar Sabie. 2025. 17 Years of Fintech for Financial Inclusion: A Systematic Review and Critical Value Analysis. ACM Conference On Computer-Supported Cooperative Work And Social Computing (ACM CSCW)
- **Priyanka Verma**, Mohammad R. Rifat, Samar Sabie. 2025. Maintenance, Cooperation, and Sustainability: Unpacking the Data Practices of Housing Cooperatives. ACM Conference on Human Factors in Computing Systems (ACM CHI)

SKILLS

ML/ AI	Prompt Engineering, Responsible AI evaluation, TensorFlow, Keras, Data Mining, ML Algorithms
Programming	Python, R, C++, C, PHP, Ruby, HTML, CSS, JavaScript, Unix and Bash script
Tools	MySQL, Git, VSCode, MATLAB, Node.js/ Flask, Docker, cplex, Qualtrics
Packages	Pandas, Gensim, NumPy, scikit-learn, OpenCV, SciPy, Matplotlib
Soft	Project Management, Presentations, Scientific writing, Cross- functional and -cultural Collaborations

AWARDS & ACHIEVEMENTS

- [Publication featured](#) in ACM Showcase on Kudos- world's largest open library to simplify peer-reviewed research
- 1 of 15 Schwartz Reisman Institute for Technology and Society (SRI) Graduate Fellow 2025-26
- 1 of 4 recipients of the Inclusive Excellence Research Grant by the Faculty of Information (2025)
- Sustainable Development Goals@UofT Student Award (2024)
- Faculty of Information PhD Enhancement Award (2024)
- School of Cities Graduate Research Fellowship (2023- 2024)
- Talent Bursary to present at Upper Bound Conference, Alberta Machine Intelligence Institute (AMII) (2023)

KEY COURSES

- Methods of Applied Statistics II: generalized linear models, Bayesian inference, advanced regression ([repository](#))
- Computational Social Science: Experiments, Observations, methods for big data research ([website](#))
- Methods for Algorithmic FATE (Fairness, Accountability, Transparency, Ethics) Research ([articles](#))